Data science portfolio

Andrei Chekunov

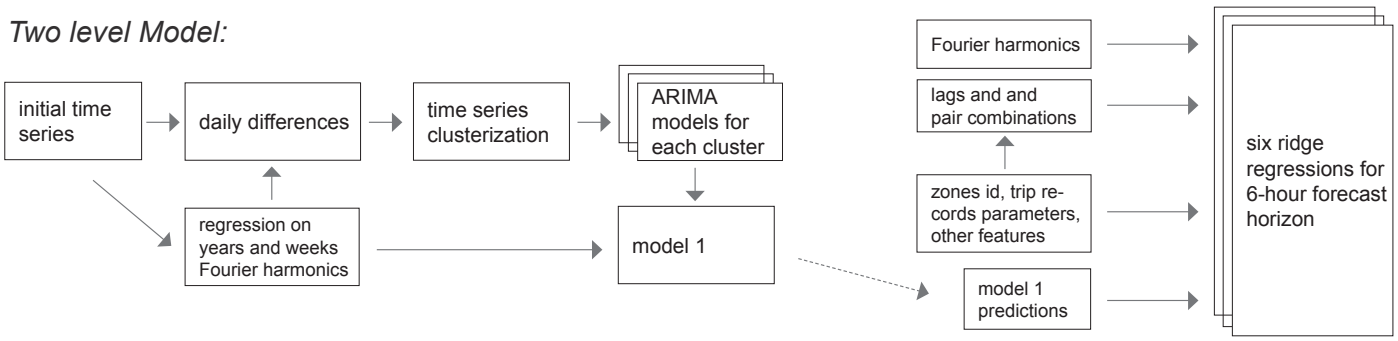
Sort-run prediction of yellow taxi trips

The final Project of ML and Data Analysis Specialisation form Yandex & Moscow Institute of Physics and Technology

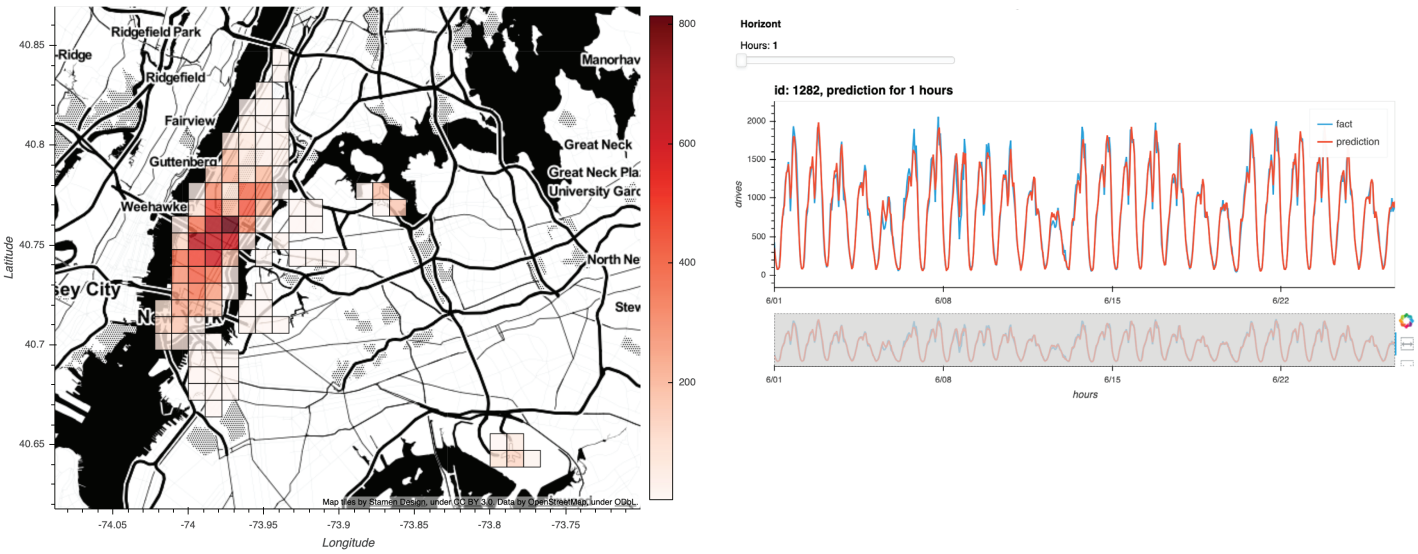
Problem: Multiple sort-run forecast of yellow taxi trips from high demand zones of New York

Data description: Yellow taxi trip records for more than 3 years: pick-up and drop-off dates/times, pick-up and drop-off locations, trip distances, rate types, etc.
https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page

Two level Model:



Results representation: An interactive widget with possibility for selection of each zone forecast for each hour of six-hours horizon



Predict Future Sales

The final project of "How to Win a Data Science Competition" Course in the Advanced ML Specialisation from HSE

Problem: Monthly sale prediction for one of the largest Russian software companies - 1C
https://www.kaggle.com/c/competitive-data-science-predict-future-sales/overview

Result: ~78th out of about 12000 teams on Kaggle, rmse: 0.839
(naïve benchmark: 1.16, the best score: 0.75)

Data: tree year daily transactions for about 60 shops / 21000 items; textual item description

Solution: https://github.com/olip78/Predict_Future_Sales/

Intruder Detection through Webpage Session Tracking

The final Project of ML and Data Analysis Specialisation form Yandex & Moscow Institute of Physics and Technology

Problem: Identification of a specific user by tracking logs of attended Web pages
https://www.kaggle.com/c/catch-me-if-you-can-intruder-detection-through-webpage-session-tracking2/

Result: ~10th place out of about 4700 teams on Kaggle; 0.972 roc-auc
(benchmark: simple BOW on sessions: 0.916, the best score: 0.985)

Data: Logs (site name, time of visit) of about, 1500 users for 16 months

Solution: https://github.com/olip78/Catch-Me-If-You-Can

Other projects

The final project of “Bayesian Methods for Machine Learning” Course in the Advanced ML Specialisation from HSE

Facial Composites. Bayesian optimization of a similarity function between a person’s face in witness’s memory and a face reconstructed from the current point of latent space of a pertained Variational Autoencoder

The final project of “Introduction to Deep Learning” Course in the Advanced ML Specialisation from HSE

Image Captioning. An ensemble of vision deep learning CNN and language generating RNN

The final project of “Natural Language Processing” Course in the Advanced ML Specialisation from HSE

StackOverflow assistant bot, a conversational assisting with search on StackOverflow

Final Projects of ML and Data Analysis Specialisation form Yandex & Moscow Institute of Physics and Technology

Customer churn prediction on blind data of a telecommunication company

Certificates / Courses

Advanced Machine Learning Specialisation, Higher School of Economics, Moscow
https://www.coursera.org/specializations/aml

5 / 7 Courses:

Introduction to Deep Learning
How to Win a Data Science Competition: Learn from Top Kagglers (with Honors)
Bayesian Methods for Machine Learning (with Honors)
Addressing Large Hadron Collider Challenges by Machine Learning
Natural Language Processing

Machine Learning and Data Analysis Specialisation, Moscow Institute of Physics and Technology & Yandex
https://www.coursera.org/specializations/machine-learning-data-analysis

6 Course Specialisation:

Math and Python,
Supervised Learning,
Unsupervised Learning and Data Visualisation,
Statistics for Data Science,
Practical Data Analysis,
Final Project

Certificate

Deep Learning Specialization, Deep Learning.AI
https://www.coursera.org/specializations/deep-learning
1 / 5 Courses (in progress)

Industrial Machine Learning Specialisation, Higher School of Economics
https://www.coursera.org/specializations/aml

2 / 3 Courses:

Big Data Processing (Spark, Hadoop, Map Reduce, Mongo DB, SQL, Bash)
Machine Learning on Big Data (ETL, Vowpal Wabbit, Spark, H2O, Feature Hashing, Scaling Computations, Distributed Learning, Transfer Learning etc.)

Other Courses

Data Visualization and Communication with Tableau, Duke University
Databases and SQL for Data Science with Python, IBM